使用 REST API 管理 ONTAP

ONTAP Automation

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使用 REST API 管理 ONTAP

使用 ONTAP REST API 管理 EMS 事件通知

您可以使用 ONTAP REST API 配置向您发送事件管理系统（ EMS ）事件通知的重要性，以及检索所有 EMS 消息或具有特定属性的特定消息。您可以使用 REST API 帮助自动执行与 EMS 相关的任务，与其他处理 EMS 消息的方法相比，可以节省时间。

以下示例显示了如何使用 ONTAP REST API 处理 EMS 消息。在每个示例中，请务必在所示位置用方括号 <> 提供信息值。您也可以引用示例 "Python 脚本" 这展示了如何自动化执行许多与 EMS 相关的活动。

查看特定事件日志

使用 `GET /support/EMS/Events` API 调用，您可以检索特定事件消息，例如最新消息，包含特定文本的消息或具有特定严重性的消息。

示例请求：检索最新消息

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/support/ems/events?fields=message.name&max_recor
ds=1'
```

响应示例


```
"records": [
    {
      "node": {
        "name": "malha-vsim1",
        "uuid": "da4f9e62-9de3-11ec-976a-005056b369de",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/da4f9e62-9de3-11ec-976a-
005056b369de"
          }
        }
      },
      "index": 6260,
      "message": {
        "name": "netinet6.rtr.high.mtu"
      },
      "_links": {
        "self": {
          "href": "/api/support/ems/events/malha-vsim1/6260"
        }
      }
    }
  ],
  "num_records": 1,
  "_links": {
    "self": {
      "href": "/api/support/ems/events?fields=message.name&max_records=1"
    },
    "next": {
      "href": "/api/support/ems/events?start.keytime=2022-03-
22T10%3A28%3A29-04%3A00&start.node.name=malha-
vsim1&start.index=6260&fields=message.name&max_records=1"
    }
  }
}
```


示例请求：检索包含特定文本和严重性的消息

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/support/ems/events?log_message=*disk*&message.se
verity=alert'
```

响应示例


```
"records": [
    {
      "node": {
        "name": "malha-vsim1",
        "uuid": "da4f9e62-9de3-11ec-976a-005056b369de",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/da4f9e62-9de3-11ec-976a-
005056b369de"
          }
        }
      },
      "index": 4602,
      "time": "2022-03-18T06:37:46-04:00",
      "message": {
        "severity": "alert",
        "name": "raid.autoPart.disabled"
      },
      "log_message": "raid.autoPart.disabled: Disk auto-partitioning is
disabled on this system: the system needs a minimum of 4 usable internal
hard disks.",
      "_links": {
        "self": {
          "href": "/api/support/ems/events/malha-vsim1/4602"
        }
      }
    }
  ],
  "num_records": 1,
  "_links": {
    "self": {
      "href":
"/api/support/ems/events?log_message=*disk*&message.severity=alert&max_rec
ords=1"
    },
    "next": {
      "href": "/api/support/ems/events?start.keytime=2022-03-
18T06%3A37%3A46-04%3A00&start.node.name=malha-
vsim1&start.index=4602&log_message=*disk*&message.severity=alert"
    }
  }
}
```

查看现有 EMS 配置

使用 `get /support/EMS/` API 调用，您可以检索有关现有 EMS 通知配置的信息。

请求示例

```
curl -X GET -u admin:<PASSWORD> -k 'https://<IP_ADDRESS>/api/support/ems/'
```

响应示例

```
{
  "proxy_url": "https://proxyserver.mycompany.com",
  "proxy_user": "proxy_user",
  "mail_server": "mail@mycompany.com",
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "pubsub_enabled": "1",
  "mail_from": "administrator@mycompany.com"
}
```

创建新的 EMS 通知配置

您可以同时使用多个 API 调用来创建新的 EMS 配置，以自定义接收消息的方式和时间。

步骤

1. 配置系统范围的电子邮件设置。例如：

   ```
   curl -X PATCH -u admin:<PASSWORD> -k
   'https://<IP_ADDRESS>/api/support/ems?mail_from=administrator@mycompany.
   com&mail_server=mail@mycompany.com'
   ```

2. 定义使用特定规则与特定消息匹配的筛选器。例如：

   ```
   curl -u admin:<PASSWORD> -X POST -d '{"name": "test-filter",
   "rules.type": ["include"], "rules.message_criteria.severities":
   ["emergency"]}' -k 'https://<IP_ADDRESS>/api/support/ems/filters/'
   ```

3. 为消息创建目标。例如：

```
curl -u admin:<PASSWORD> -X POST -d '{"name": "test-destination",
"type": "email", "destination": "administrator@mycompany.com",
"filters.name": ["important-events"]}' -k
'https://<IP_ADDRESS>/api/support/ems/destinations/'
```

ONTAP REST API 与 ONTAP 命令行界面命令

与 ONTAP 命令行界面相比， ONTAP REST API 可以使用更少的命令自动执行许多任务。例如，您可以使用一个 POST API 方法来创建筛选器，而不是使用多个 CLI 命令。下表显示了完成常见 EMS 任务与相应的 REST API 调用所需的命令行界面命令：

ONTAP REST API	ONTAP 命令行界面
`GET /support/EMS`	`event config show`
`POST /support/EMS/destinations`	1. 事件通知目标创建 2. 事件通知创建
`GET /support/EMS/Events`	事件日志显示
`POST /support/EMS/filters`	1. `event filter create -filter-name <FilterName>` 2. `event filter rule add -filter-name <FilterName>`

相关信息

- "ONTAP REST API EMS Python 脚本示例"
- "ONTAP REST API： 自动通知高严重性事件"

使用 ONTAP REST API 管理 DACL 和 SACL 文件权限

ONTAP 使用系统访问控制列表（ SACL ）和随机访问控制列表（ DACL ）为文件对象分配权限。从 ONTAP 9.1.1 开始， ONTAP REST API 包括用于为文件分配 SACL 和 DACL 权限以及自动执行文件安全权限的端点。（您可以了解有关在 ONTAP 9.9.1 之前自动执行 SACL 和 DACL 权限的选项的更多信息 "此处"）。

从 ONTAP 9.1.1 开始，您可以使用单个 REST API 调用来代替多个 CLI 命令或 ONTAPI 调用。以下示例显示了如何使用 ONTAP REST API 处理文件权限。在每个示例中，请务必在所示位置用方括号 <> 提供信息值。

您也可以引用示例 "Python 脚本" 该演示了如何自动执行许多与 SACL 和 DACL 相关的活动。

查看有效权限

使用 `get /protocols/file-security/effective-permissions/` API 调用，您可以检索特定文件或目

录的当前权限。

请求示例：

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/protocols/file-security/effective-
permissions/cf5f271a-1beb-11ea-8fad-
005056bb645e/administrator/windows/%2F?share.name=sh1&return_records=true'
```

响应示例

```
{
  "svm": {
    "uuid": "cf5f271a-1beb-11ea-8fad-005056bb645e",
    "name": "vs1"
  },
  "user": "administrator",
  "type": "windows",
  "path": "/",
  "share": {
    "path": "/"
  },
  "file_permission": [
    "read",
    "write",
    "append",
    "read_ea",
    "write_ea",
    "execute",
    "delete_child",
    "read_attributes",
    "write_attributes",
    "delete",
    "read_control",
    "write_dac",
    "write_owner",
    "synchronize",
    "system_security"
  ],
  "share_permission": [
    "read",
    "read_ea",
    "execute",
    "read_attributes",
    "read_control",
    "synchronize"
  ]
}
```

查看所有审核信息

使用 `get /protocols/file-security/permissions/` API 调用，您可以检索特定文件或目录的所有审核信息。

请求示例:

```
curl -X GET -u admin:<PASSWORD> -k
'https://<IP_ADDRESS>/api/protocols/file-security/permissions/9479099d-
5b9f-11eb-9c4e-0050568e8682/%2Fparent'
```

响应示例


```
{
  "svm": {
    "uuid": "9479099d-5b9f-11eb-9c4e-0050568e8682",
    "name": "vs1"
  },
  "path": "/parent",
  "owner": "BUILTIN\\Administrators",
  "group": "BUILTIN\\Administrators",
  "control_flags": "0x8014",
  "acls": [
    {
      "user": "BUILTIN\\Administrators",
      "access": "access_allow",
      "apply_to": {
        "files": true,
        "sub_folders": true,
        "this_folder": true
      },
      "advanced_rights": {
        "append_data": true,
        "delete": true,
        "delete_child": true,
        "execute_file": true,
        "full_control": true,
        "read_attr": true,
        "read_data": true,
        "read_ea": true,
        "read_perm": true,
        "write_attr": true,
        "write_data": true,
        "write_ea": true,
        "write_owner": true,
        "synchronize": true,
        "write_perm": true
      },
      "access_control": "file_directory"
    },
    {
```

```
      "user": "BUILTIN\\Users",
      "access": "access_allow",
      "apply_to": {
        "files": true,
        "sub_folders": true,
        "this_folder": true
      },
      "advanced_rights": {
        "append_data": true,
        "delete": true,
        "delete_child": true,
        "execute_file": true,
        "full_control": true,
        "read_attr": true,
        "read_data": true,
        "read_ea": true,
        "read_perm": true,
        "write_attr": true,
        "write_data": true,
        "write_ea": true,
        "write_owner": true,
        "synchronize": true,
        "write_perm": true
      },
      "access_control": "file_directory"
    }
  ],
  "inode": 64,
  "security_style": "mixed",
  "effective_style": "ntfs",
  "dos_attributes": "10",
  "text_dos_attr": "----D---",
  "user_id": "0",
  "group_id": "0",
  "mode_bits": 777,
  "text_mode_bits": "rwxrwxrwx"
}
```


应用新权限

使用 `POST /protocols/file-security/permissions/` API 调用，您可以将新的安全描述符应用于文件或目录。

请求示例

```
curl -u admin:<PASSWORD> -X POST -d '{ \"acls\": [ { \"access\":
\"access_allow\", \"advanced_rights\": { \"append_data\": true,
\"delete\": true, \"delete_child\": true, \"execute_file\": true,
\"full_control\": true, \"read_attr\": true, \"read_data\": true,
\"read_ea\": true, \"read_perm\": true, \"write_attr\": true,
\"write_data\": true, \"write_ea\": true, \"write_owner\": true,
\"write_perm\": true }, \"apply_to\": { \"files\": true, \"sub_folders\":
true, \"this_folder\": true }, \"user\": \"administrator\" } ],
\"control_flags\": \"32788\", \"group\": \"S-1-5-21-2233347455-2266964949-
1780268902-69700\", \"ignore_paths\": [ \"/parent/child2\" ], \"owner\":
\"S-1-5-21-2233347455-2266964949-1780268902-69304\", \"propagation_mode\":
\"propagate\"}' -k 'https://<IP_ADDRESS>/api/protocols/file-
security/permissions/9479099d-5b9f-11eb-9c4e-
0050568e8682/%2Fparent?return_timeout=0'
```

响应示例

```
{
  "job": {
    "uuid": "3015c294-5bbc-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/3015c294-5bbc-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

更新安全描述符信息

使用 `patch /protocols/file-security/permissions/` API 调用，您可以更新文件或目录的特定安全描述符信息，例如主所有者，组或控制标志。

请求示例

```
curl -u admin:<PASSWORD> -X PATCH -d '{ \"control_flags\": \"32788\",
\"group\": \"everyone\", \"owner\": \"user1\"}' -k
'https://<IP_ADDRESS>/api/protocols/file-security/permissions/9479099d-
5b9f-11eb-9c4e-0050568e8682/%2Fparent?return_timeout=0'
```

响应示例

```
{
  "job": {
    "uuid": "6f89e612-5bbd-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/6f89e612-5bbd-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

删除现有 SACL/DACL 访问控制条目（ACE）

使用 `Delete /protocols/file-security/permissions/` API 调用，您可以从目录的文件中删除现有 ACE 。此示例将此更改传播到任何子对象。

请求示例

```
curl -u admin:<PASSWORD> -X DELETE -d '{ \"access\": \"access_allow\",
\"apply_to\": { \"files\": true, \"sub_folders\": true, \"this_folder\":
true }, \"ignore_paths\": [ \"/parent/child2\" ], \"propagation_mode\":
\"propagate\"}' -k 'https://<IP_ADDRESS>/api/protocols/file-
security/permissions/9479099d-5b9f-11eb-9c4e-
0050568e8682/%2Fparent/acl/himanshu?return_timeout=0'
```

响应示例

```
{
  "job": {
    "uuid": "e5683b61-5bbf-11eb-9c4e-0050568e8682",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/e5683b61-5bbf-11eb-9c4e-0050568e8682"
      }
    }
  }
}
```

ONTAP REST API 与 ONTAP 命令行界面命令

与 ONTAP 命令行界面相比， ONTAP REST API 可以使用更少的命令自动执行许多任务。例如，您可以使用一种 POST API 方法来修改文件的安全描述符，而不是使用多个 CLI 命令。下表显示了完成常见文件系统权限任务与相应的 REST API 调用所需的命令行界面命令：

ONTAP REST API	ONTAP 命令行界面
`get /protocols/file-security/effective-permissions/`	`vserver security file-directory show-effective-permissions`
`POST /protocols/file-security/permissions/`	1. `vserver security file-directory ntfs create` 2. `vserver security file-directory ntfs dacl add` 3. `vserver security file-directory ntfs sacl add` 4. `vserver security file-directory policy create` 5. `vserver security file-directory policy task add` 6. `Vserver security file-directory apply`
`patch /protocols/file-security/permissions/`	`vserver security file-directory ntfs modify`
`delete /protocols/file-security/permissions/`	1. `vserver security file-directory ntfs dacl remove` 2. `vserver security file-directory ntfs sacl remove`

相关信息

- "ONTAP REST API DACL/SACL 权限示例 Python 脚本"
- "使用 ONTAP REST API 简化文件安全权限的管理"
- "将专用命令行界面直通与 ONTAP REST API 结合使用（适用于 ONTAP 9.1.1 之前的版本）"

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